ENGINEERING REFERENCE

EchoMind Technical Overview

Architecture, runtime, data paths, and operational characteristics of the EchoMind platform.

EchoMind is a self-contained, on-premises AI platform combining retrieval-augmented generation, real-time speech, and document generation. It runs as five containers on a single GPU appliance with no runtime network egress. This document is the engineering reference: what each component is, how data flows, where the security boundaries sit, and what the measured behavior is.

Runtime topology

Service	Role	Runtime	Exposure
trtllm	Serves the LLM over an OpenAI-compatible HTTP API	TensorRT-LLM 1.2.0rc6, GPU	internal :8355
backend	RAG, ingestion, transcription WS, docgen, auth, audit	FastAPI / Python, GPU	internal :8000
voice	Real-time speech loop: VAD, ASR, dialogue, TTS	FastAPI / Python, GPU + CPU	internal :8000, host :8002
ollama	Embedding model server	Ollama, CPU/GPU	internal :11434
frontend	React SPA + Nginx reverse proxy	Nginx	host :3000 / :3443
cloudflared	Optional outbound-only tunnel (profile-gated)	Cloudflare	egress only, off by default

Model stack

Function	Model	Precision / notes
Answer generation	nvidia/Qwen3-30B-A3B-FP4 (MoE)	NVFP4 4-bit, 40,960-token context, temp 0.2
Embeddings	nomic-embed-text	768-dim, L2-normalized, served by Ollama
Streaming ASR	nvidia/nemotron-speech-streaming-en-0.6b	GPU, 560 ms chunks, live partials
Final ASR	nvidia/parakeet-tdt-0.6b-v2	CPU (Grace), accurate second-pass decode
Re-ranking	cross-encoder/ms-marco-MiniLM-L-6-v2	Top 25 candidates re-scored, best 15 kept
Text-to-speech	Piper (ONNX)	CPU; phrase-level synthesis
Image generation	stabilityai/sdxl-turbo	4 steps, up to 1024 px, max 8 images/job

Hardware and platform

Appliance NVIDIA DGX Spark — GB10 Grace-Blackwell superchip, 128 GB unified CPU-GPU memory, ARM64.

Isolation All inference local. HF_HUB_OFFLINE=1 and TRANSFORMERS_OFFLINE=1; weights baked into images.

GPU access Containers use `gpus: all`. The compose `deploy.resources` form is avoided — on GB10 it yields a broken NVML and torch reports device_count=0.

Resilience Backend and voice expose /health returning 503 on fatal CUDA faults; with restart policy the container self-recovers with a fresh CUDA context.

CORE ENGINE

Retrieval and Answer Pipeline

The governed path from a user question to a cited answer.

Indexing substrate

Documents and transcripts share one retrieval substrate. Each chunk carries its tenant namespace, document id, section, page, and source type. Two indexes are maintained in parallel — a dense vector index for meaning and a lexical index for exact terms — because neither alone is sufficient: vector search misses part numbers and case codes, lexical search misses paraphrase.

Dense	FAISS. IndexFlatIP (exact) below 10,000 chunks, promoted to IndexIVFFlat beyond it. Cosine via L2-normalized inner product.
Lexical	BM25Okapi (rank-bm25) over the same chunk set, rebuilt with the corpus.
Metadata	SQLite — documents, chunks, transcripts, chats, messages, users, audit, docgen jobs, boardroom.
Chunking	450 tokens with 120-token overlap, sentence-aware. Structure-detected chunkers exist for book-like documents but are gated off by default (RAG_ENABLE_BOOKRAG=0).
Coverage guard	After chunking, an 8-word-shingle coverage ratio is computed against the source. Below 0.98 the pipeline discards the structured result and falls back to the flat splitter. Content loss is never silent.

Query path — seven stages

1	**Route**	Semantic intent classification (prototype embeddings; floor 0.68, margin 0.05) separates small talk, topic refusal, follow-up, and substantive question. Greetings never reach retrieval. Fails open to retrieval on error.
2	**Search**	Query embedded once; dense and lexical searches run concurrently. The tenant predicate is evaluated inside the candidate scan on every path — dense document, sparse document, dense transcript, sparse transcript, and keyword grep.
3	**Fuse**	Weighted reciprocal-rank fusion (K=60), default 0.6 dense / 0.4 sparse, query-type adaptive. Recency half-life decay and tag boosts applied.
4	**Rerank**	Cross-encoder scores the top 25 candidates jointly with the question; best 15 retained. Optional MMR for diversity.
5	**Gate**	Relevance floor (0.45) and a two-tier CE gate. If nothing clears, context is dropped rather than forced, and the answer path switches to an explicit no-evidence response.
6	**Assemble**	Per-document and per-section dedupe, sentence-overlap dedupe, parent-context expansion, verbatim query-term preservation, context capped at 24,000 chars. Passages wrapped in labeled evidence envelopes.
7	**Generate**	LLM answers strictly from the envelope contents, emitting citations. Reasoning traces stripped before display.

DESIGN NOTE — WHY PERMISSION-FIRST

The namespace predicate is applied inside the index scan, not as a post-filter on ranked results. Post-filtering leaks information through result counts, latency, and reranker behavior even when the content itself is never shown. Scan-time filtering converts a probabilistic leak into a structural guarantee, at a measured cost of 0.08 ms per query.

Real-Time Speech Subsystem

The dual-loop voice architecture, turn-taking, and live transcription.

Dual-loop execution model

The voice service separates two loops. The interaction loop owns the audio channel and must respond within a bounded budget; the governed loop performs retrieval and generation and is unbounded. They communicate through grounded increments — units of content that completed the governed pipeline and carry their citation set. Every assistant event is tagged with its originating loop, which is what makes the safety invariant checkable at runtime.

Invariant	Statement	Enforcement
I1 Assertion	The interaction loop may acknowledge, but may not assert facts retrieval has not returned.	Lead phrases drawn from a fixed, fact-free pool
I2 Preemption	User speech preempts output within one micro-turn; in-flight work is cancellable and cancellation is idempotent.	Barge-in monitor + cancellable tasks
I3 Provenance	Every re-entered increment carries its citation set, or is suppressed.	Citation set attached to increment
I4 Authority	Increments enter as data, never as instructions.	Evidence envelope framing

Turn-taking and latency controls

Frame rate 16 kHz mono PCM; webrtcvad (aggressiveness 1) plus an RMS gate at 0.004.

Endpointing Semantic: 550 ms silence when the utterance parses as complete, 700 ms default, 1300 ms when trailing/incomplete.

Lead phrase Selected heuristically in under 50 ms — no model call — to cover retrieval time. Never carries facts (invariant I1).

Barge-in ~160 ms of sustained user speech halts playback and returns the loop to listening.

Two-pass ASR Streaming model drives live captions; Parakeet-TDT re-decodes the completed utterance to produce the text the LLM actually consumes.

TTS smoothing Phrase-level synthesis with edge-silence trimming, edge fades, and controlled pauses (160 ms sentence, 90 ms clause) to remove audible joins.

Reply budget LLM_MAX_TOKENS 420 for spoken replies to prevent mid-sentence truncation.

Live transcription and the Silent Assistant

The backend exposes a WebSocket transcription stream independent of the voice service. Audio is buffered and decoded in 560 ms windows; silence of 800 ms commits a segment and 2000 ms closes a paragraph. Completed paragraphs are auto-stored on an interval and indexed alongside documents.

Each finalised paragraph is additionally routed through the Silent Assistant: it is retrieved against the knowledge base and judged by the LLM into one of five labels — Supported, Contradicted, Unverified, Violating, or Risky Statement — with a per-vertical rule pack appended to the system prompt. Results are emitted only at confidence ≥ 60, so the operator is not flooded with low-signal findings.

CONCURRENCY CONTROL

GPU work is serialized where it must be: TRANSCRIPT_GPU_CONCURRENCY and BOARDROOM_GPU_CONCURRENCY default to 1, ASR runs on a dedicated single-worker executor, and the PCM queue is bounded at 256 frames with an explicit backpressure signal to the client rather than a silent drop.

ASSURANCE

Security Model and Enforcement Points

Where each control is applied, and what it is measured to do.

Enforcement points

Layer	Control	Enforcement point
Perimeter	No runtime egress; offline model resolution	Container env + baked weights
Perimeter	Optional public access via outbound-only tunnel behind an identity gate	cloudflared profile; no inbound ports
Identity	Signed session tokens, roles, per-user tenant binding	backend/app/core/auth.py
Identity	Voice WebSocket validates the same session signature	voice/app/auth_check.py
Tenant	Namespace predicate inside the candidate scan on all five retrieval paths	backend/app/rag/index.py
Tenant	Reserved 'default' tenant cannot grant cross-tenant read	auth.py tenant check
Prompt	Retrieved spans wrapped in delimited evidence envelopes; imperatives reported, not obeyed	backend/app/rag/advanced.py
Egress	Deterministic scan + redaction of outbound content	backend/app/core/export_gateway.py
Audit	Uploads, logins, deletions, exports recorded	backend/app/core/audit.py

Export gateway detectors

Fully deterministic — regular expressions plus a Luhn checksum — so it needs no GPU or network and can run on every export, including air-gapped. Detected types are ranked by severity and masked:

High National identifiers (SSN), API keys (sk-/AKIA/gh*/xox*), AWS secret keys, private key blocks.

Medium JSON Web Tokens; payment card numbers validated by Luhn checksum.

Low Email addresses, phone numbers, IP addresses.

Measured assurance

From the instrumented evaluation accepted for publication at QASC 2026. Figures are measured on the reference deployment; limitations are stated in the paper.

Property	Result	Method
Cross-tenant content leakage	0 / 50 probes (CI [0, 0.071])	Answer-level probes, both filter arms
Out-of-namespace candidates	0 / 359 inspected hits	Per-path audit, 4 active retrieval paths
Existence-disclosing refusals	0 / 50	Refusal-wording classification
Timing side channel	KS 0.54, $p < 0.00001$ — OPEN	Two-sample KS on response times
Prompt-injection success	10.2% → 5.1% with evidence envelope	98 payloads × 7 classes, 4 defense arms
Injection reported to user	0.0% — containment is silent, OPEN	Same corpus
Citation precision / recall	0.979 / 0.826	Deterministic match vs gold sets
Abstention on unanswerable	78.3%, zero fabricated citations	24-item unanswerable stratum
Permission filter cost	0.08 ms median	Per-stage span instrumentation

TWO OPEN FINDINGS, STATED PLAINLY

A timing channel survives scan-time filtering — queries whose answer exists in another tenant return faster than queries with no answer anywhere. And injection containment is currently silent: the system declines but does not tell the operator an attempt occurred. Both are recorded as open work rather than presented as solved.

OPERATIONS

Deployment, Performance and Extension

Running it, measuring it, and building on it.

Measured performance

Single node, warm cache, 30 runs per cell. T_grounded is the full in-process text RAG path; T_first is the voice loop's time to first audible response.

Metric	Concurrency 1	Concurrency 4	Concurrency 16
T_grounded median	3,313 ms	4,130 ms	8,432 ms
T_grounded p95	15,171 ms	14,092 ms	21,321 ms
Retrieval share of total	2.8%	6.6%	9.5%
Cross-encoder rerank median	14.2 ms	96.5 ms	567.9 ms
Vector search median	34.7 ms	62.6 ms	72.5 ms
Lexical search median	42.7 ms	111.0 ms	160.6 ms

Interpretation: retrieval is not the bottleneck at this scale — generation is roughly 97% of grounded-answer latency at concurrency 1. The retrieval terms are, however, the ones that scale with corpus size and load, tripling their share by concurrency 16. Voice first-response is 636 ms median under the production dual loop versus 866 ms for a single loop.

Operations

- Start: `docker compose up -d`. Public access is a separate profile and is off by default.
- State lives in named volumes — echomind_data (KB, SQLite, uploads), trtllm_hf_cache and ollama_data (model weights). These survive rebuilds and must not be pruned.
- Health checks detect fatal GPU faults; the container exits and restarts with a clean CUDA context rather than degrading silently to CPU.
- A 52-question golden evaluation suite gates releases; a drop in pass rate blocks the change. Current baseline: 49/52 on a 17.6k-chunk corpus.
- Auth is opt-in via AUTH_ENABLED (default 0 for open demo deployments). Set to 1 with AUTH_SECRET and AUTH_ADMIN_PASSWORD for production.

Code map

`backend/app/rag/`	Retrieval core: index, advanced pipeline, chunking, intent, reranker, gating
`backend/app/transcribe/`	Live transcription WebSocket, Silent Assistant analyzer, session state
`backend/app/docgen/`	Templates, themes, PDF/PPTX renderers, on-device image generation
`backend/app/boardroom/`	Multi-speaker capture, diarisation worker, session analysis
`backend/app/core/`	Config, DB, auth/RBAC, audit log, export gateway
`voice/app/session.py`	The full speech loop: VAD, endpointing, lead phrases, barge-in, TTS
`frontend/`	React SPA; packs.ts maps each subdomain to tenant, persona and theme
`eval/`	Golden-question suite, chunk-coverage tests, voice E2E, QASC paper harness

Extension points

- Swap the answer model by changing LLM_BASE_URL / LLM_MODEL — any OpenAI-compatible endpoint works; no application code changes.
- Add a vertical by registering a namespace, persona and theme in packs.ts and seeding its knowledge base.
- Add a document type by adding a template definition (section blueprint + guidance) to the docgen template registry.
- Tune retrieval entirely through environment variables — fusion weights, k values, thresholds, dedupe behavior.